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PRYMNESIUM PARVUM: THE TEXAS EXPERIENCE[1]

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ABSTRACT: Golden alga *Prymnesium parvum* was first identified in Texas during a fish kill investigation on the Pecos River in 1985. Since then golden alga kills occurred sporadically in a variety of waters in the western part of the state until 2001 when the alga became endemic in the Brazos, Canadian, Colorado, Red, and Rio Grande river systems, including the water supplies of two public fish hatcheries, the Possum Kingdom and Dundee state fish hatcheries. The increasing area adversely affected by the alga and frequent massive fish kills heightened public and political awareness and concerns regarding the ecological and economic impacts of *P. parvum* blooms. The Texas Parks and Wildlife Department (TPWD), the wildlife conservation agency of the state, responded to these concerns with a program to assess the ecological and economic impacts and to develop management options. To date 33 water bodies have been affected and losses are conservatively estimated at 34 million fish valued at US$13 million. Several sport fisheries, including smallmouth bass *Micropterus dolomieu*, striped bass *Morone saxatilis*, channel catfish *Ictalurus punctatus*, and blue catfish *Ictalurus furcatus*, have been severely affected. Additionally, 26 imperiled fish species occur in the affected water basins and some have been adversely affected. Economic losses associated with reduced fishing and other water-based recreational activities appear considerable. The combined economic losses to three counties (Palo Pinto, Stephens, and Young) surrounding Possum Kingdom reservoir for 2001 and 2003 were estimated at US$2.8 million and US$1.1 million, respectively. This paper describes how the TPWD responded to public and political concerns relative to the emergence of golden alga, its harmful effects to fisheries, and its historic and current statewide distribution.

(KEY TERMS: algae; aquatic ecology; environmental impacts; fish; toxicology; water resources economics.)

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INTRODUCTION

The haptophyte *Prymnesium parvum* (Figure 1) is a toxin-producing alga (Dafni and Shilo, 1966; Meldahl *et al.*, 1995) of considerable ecological and economic importance for causing extensive mortalities in fish and other gill-breathing organisms (Holdway *et al.*, 1978; Kaartvedt *et al.*, 1991; Guo *et al.*, 1996). First described in 1935 in England by Carter (1937), *P. parvum* (golden alga) is considered ubiquitous (Bruslé, 1995). This alga and its associated fish kills have been reported from at least 14 countries and from every continent except Antarctica (Farrow,

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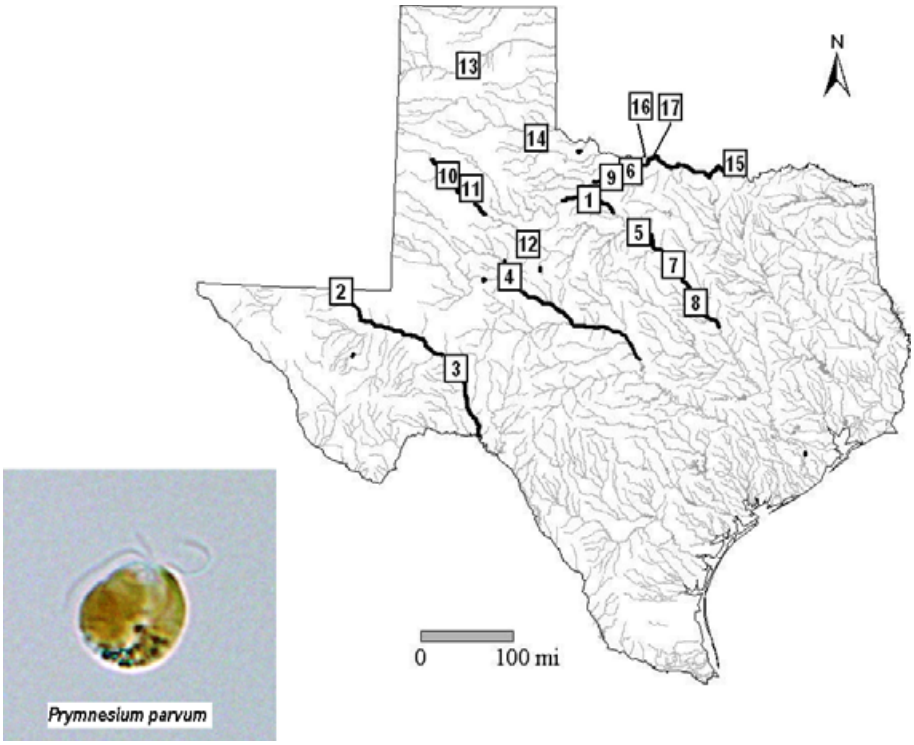

FIGURE 1. Locations of Significant Fish Kills in Texas Related to *Prymnesium parvum*. Water-body names for sites can be found in Table 1. Numbers are indicated in chronological order when golden alga was first suspected or confirmed at the sites (1980-2008). Blackened areas indicate current range of golden alga in the state.

1969; Larsen and Bryant, 1998). Although *P. parvum* is most often found in marine and estuarine waters (Shilo and Aschner, 1953), it can bloom and cause toxicity in brackish inland waters (Linam *et al.*, 1991). The toxins produced by *P. parvum* have cytotoxic, hemolytic, neurotoxic, and ichtyotoxic activities (Yariv and Hestrin, 1961; Parnas, 1963; Paster, 1973); however, massive toxicity-related mortalities in nature have been reported only in gill-breathing animals such as fish and tadpoles (Shilo and Aschner, 1953; Paster, 1973). Adverse effects of these toxins on higher vertebrates, including humans, have not been documented (Paster, 1973; Lindholm *et al.*, 1999).

Although *P. parvum* has caused massive mortalities in propagated and wild fish worldwide, it has been most active in recent years in the Baltic region, especially Norway and Finland (Lindholm *et al.*, 1999), and the United States (U.S.) (Sager *et al.*, 2008). In the U.S., golden alga has been reported from 16 states (Sager *et al.*, 2008), though bloom formation and related fish kills have been most frequent, extensive, and severe in Texas since 2001 resulting in increased public and political interests in this alga and its impacts. Consequently, the Texas Parks and Wildlife Department (TPWD) has lead efforts to manage golden alga related problems in the state. This paper describes the emergence of the golden alga problem in Texas, its impacts on fisheries and local economies, and the TPWD's efforts to manage this harmful alga and its effects.

HISTORICAL PERSPECTIVE OF AN EMERGING PROBLEM

The first confirmation of *P. parvum* in Texas resulted from investigations into a series of fish kills along the Pecos River (Figure 1) in 1985 by state authorities, the TPWD and Texas Water Commission (James and De La Cruz, 1989; Linam *et al.*, 1991). An estimated 110,000 fish died due to *P. parvum* toxicity in river waters stretching from Iraan in Pecos County to Amistad Reservoir in Val Verde County from October through November 1985. Millions of the invasive Asiatic clam *Corbicula fluminea*, which seemed particularly sensitive to toxins produced by *P. parvum*, also died (Linam *et al.*, 1991). During late October and early November of 1986, more than 500,000 fish, including state listed species-of-concern, the Pecos pupfish *Cyprinodon pecosensis*, Rio Grande darter *Etheostoma grahami*, and blue sucker *Cycleptus elongatus*, perished in the same general area as

the 1985 fish kill. Although not confirmed, fish kills occurring in the upper Brazos River in 1981-1982 and in 1985 in Red Bluff Reservoir (an impoundment of the Pecos River) are suspected to be caused by *P. parvum*, as well. More fish kills were confirmed in the following years as the range of this alga seemed to expand. Catastrophic kills involving more than 1.5 million fish and other aquatic species occurred during 1988 and 1989 in the lower Pecos River including Red Bluff Reservoir. Also during that year, *P. parvum* was confirmed in a kill near Abilene at Paint Creek, a tributary of the Brazos River where 48,000 fish were estimated to have died. In 1989, golden alga spread to another river basin as fish kills were reported in waters along the Colorado River in Coleman, Coke, Mills, Runnels, and San Saba Counties (Figure 1). By the end of the decade, TPWD documented 15 fish kills suspected or confirmed to be caused by golden alga, killing more than 2.6 million fish with an estimated value of US$2.76 million.

Through the 1990s, relatively minor fish kills attributed to *P. parvum* occurred sporadically in west Texas waters (Table 1). In comparison with the previous decade, when *P. parvum* was first identified in Texas and seemed to expand its range in a relatively short time, fish kills during the 1990s occurred less frequently with only five confirmed golden alga fish kills involving an estimated 680,000 fish.

In 2001, the situation worsened as more major water bodies experienced significant golden alga fish kills. Early that year, a kill of over 200,000 fish occurred at Possum Kingdom Reservoir, a large impoundment of the upper Brazos River. Two downstream reservoirs, Lake Granbury and Lake Whitney, also experienced *P. parvum* fish kills that year. In addition, reservoirs of the upper Colorado River, including Lake Colorado City and Moss Creek, experienced fish kills. The most significant kill occurred at Lake E.V. Spence where over two million fish died. Also in 2001, golden alga appeared in the hatchery source water, Diversion Lake, and ponds of the Dundee state fish hatchery (DSFH), a TPWD facility located along the Wichita River in the Red River Basin.

Since 2001, public waters in Texas have experienced more frequent fish kills in new areas and larger water systems than previously observed. However, it is unknown if the range of *P. parvum* is actually increasing or if TPWD staffs have improved their abilities to locate and investigate fish kills. Another

TABLE 1. Summary of Investigated Toxic Events Related to *Prymnesium parvum* in Texas Water Bodies for 1981 to 2008 (TPWD KAST PRISM data).

Basin	Water Body	Year(s) of Documented Fish Kills	Fish Kills (*n*)	Estimated Fish Loss	Estimated Economic Loss (US$)	ID (Figure 1)
Brazos	Brazos River	1981-1982*, 1988-1989, 1992, 1997-1998, 2003, 2005-2007	17	1,843,945	414,087	1
	Buffalo Springs	2003, 2006, 2008	3	10,958	22,395	11
	Granbury	2001, 2003-2007	7	8,466,951	2,156,413	7
	Lubbock City Lakes	2003, 2005-2006	6	33,598	30,780	10
	Possum Kingdom	2001-2007	12	5,473,114	1,496,992	5
	Sweetwater	2003	1	7,801	75,400	12
	Whitney	2001, 2003-2007	9	6,883,662	2,233,054	8
	Other kills	2006-2008	2	2,746	3,586	
Canadian	Meredith (Stilling Basin)	2003, 2006, 2008	2	288	1,803	13
Colorado	Colorado City	2001-2002, 2004-2007	7	68,327	317,513	4
	Colorado River	1989, 2001, 2003-2006	12	94,179	40,072	11
	E.V. Spence	2001-2006, 2008	7	2,567,610	544,330	14
	Other kills	2001-2008	11	16,846	8,531	
Red	Baylor	2003-2006, 2008	7	2,598	2,338	14
	Diversion[1]	2001-2002, 2004-2007	8	5,680,667	2,392,657	6
	Kemp	2002-2003, 2005-2008	8	15,144	51,726	9
	Texoma	2003-2004, 2006-2007	4	1,216,699	420,808	15
	Wichita	2004, 2007	3	23,363	4,799	17
	Little Wichita River	2004, 2006	2	12,448	42,861	16
	Other kills	2004-2006, 2008	10	8,689	9,549	
Rio Grande	Pecos River	1985-1986, 1993, 1995, 2005, 2007	9	438,174	2,295,538	3
	Red Bluff	1985, 1988-1989, 2003, 2006-2007	6	1,594,655	413,076	2
Totals			153	34,462,462	12,978,308	

Notes: TPWD, Texas Parks and Wildlife Department. Suspected golden alga fish kills are denoted with an asterisk (*) and all other dates are confirmed kills.
[1]Includes fish loss at the Dundee State Fish Hatchery in 2001 estimated at $426,739.

possibility is that factors conducive to toxic bloom formation are becoming more widespread. For example, total dissolved solids in the Pecos River, where the first golden alga related fish kills were confirmed, were documented to have increased significantly between 1938 and 1981 (Davis, 1987). Currently, 33 reservoirs in five river basins (Brazos, Colorado, Canadian, Red, and Rio Grande drainages) have been affected by *P. parvum* fish kills (Table 1). Additional golden alga fish kills have occurred in several community fishing lakes, streambeds, and stock ponds on private lands (Figure 1). Since 1985 TPWD has investigated over 120 golden alga fish kills, predominantly in waters in central and western Texas and from fall or late winter to late spring or early summer. An estimated 34 million fish valued at nearly US$13 million have perished due to this harmful alga. This estimate does not include loss of other aquatic species or recreational revenues to local economies.

IMPACTS TO AQUATIC RESOURCES AND LOCAL ECONOMIES

Threat to Sport Fisheries and Imperiled Fish Species

The TPWD Inland Fisheries Division is responsible for the management and protection of the diverse freshwater fisheries resources of Texas. Approximately 800 public impoundments comprising 1.7 million surface acres and over 191,000 miles of rivers and streams possess diverse fish species and other aquatic life. Observations of golden alga fish kills by TPWD suggest a progression of fish mortality beginning with abundant forage species and ending with rough species such as common carp *Cyprinus carpio*, gar *Lepisosteus* spp., and buffalo *Ictiobus* spp. (Sager *et al.*, 2008). Although all species and sizes of fish are believed to be susceptible to golden alga toxicity, the majority of fish killed have been forage species, particularly gizzard shad *Dorosoma cepedianum* and threadfin shad *Dorosoma petenense*. Because forage fish have short lifetimes and quickly restore their numbers, no long-term adverse effects to these populations have been observed. Unlike forage fish species, game fishes such as temperate bass *Morone* spp. and their hybrids, black bass *Micropterus* spp., and catfish *Ictalurus* spp., can take years to reach sexual maturity and population structures can be altered by these fish kills. Among the once thriving sport fisheries that have been adversely affected are the smallmouth bass *Micropterus dolomieu* fishery in Lake Whitney (Baird and Tibbs, 2008) and striped bass *Morone saxatilis* fishery in Possum Kingdom Reservoir (Howell and Mauk, 2007), both impoundments of the Brazos River. In some reservoirs, a variety of sport fish species have been adversely affected. Lake E.V. Spence, on the Colorado River, has experienced repeated fish kills since 2001 (Table 2) and no channel catfish *Ictalurus punctatus*, blue catfish *Ictalurus furcatus*, or striped bass have been captured in gill nets since 2002 (Farooqi and Scott, 2008). Other reservoirs experiencing dramatic declines in fisheries as a result of golden alga toxicity are Lake Colorado City (Scott and Van Zee, 2005a), Diversion Lake (Mauk and Howell, 2005), Lake Granbury (Baird and Tibbs, 2006), Lake Kemp (Howell and Mauk, 2006), Moss Creek, and Red Bluff Reservoir (Scott and Van Zee, 2005b).

Toxic waters rarely comprise entire water systems, so there are often refuge areas where fish can survive otherwise toxic blooms. Thus, despite recurring lethal conditions, some water bodies may experience a rebound in some sport fish populations through natural recruitment, restoration efforts such as fish stockings, or both. A 2007 survey at Lake Whitney indicated a promising rebound in largemouth bass *Micropterus salmoides* and white bass *Morone chrysops* populations (Baird and Tibbs, 2008). At Possum Kingdom Reservoir, populations of blue catfish, channel catfish, white bass, and largemouth bass are healthy despite repeated kills since 2001 (Howell and Mauk, 2007). Biologists recommend promoting these fisheries to re-direct angling effort and counter public misconceptions that fisheries in these reservoirs have been totally destroyed by golden alga.

There are 26 imperiled fish species (i.e., those that are federally or state listed as threatened or endangered) within the five Texas water basins known to be affected by *P. parvum*. Sixteen imperiled species occur in the Rio Grande Basin alone, such as the federally endangered Rio Grande Silvery minnow *Hybognathus amarus*, Leon Springs pupfish *Cyprinodon bovinus*, and Big Bend gambusia *Gambusia gaigei* (http://www.gis.tpwd.state.tx.us/TpwEndangeredSpecies/DesktopDefault.aspx, *accessed* August 2009). In addition, there are also several species-of-concern, which may require listing in the foreseeable future if observed trends continue. Populations of three state-listed species, the blue sucker, Pecos pupfish, and the Rio Grande darter, were affected during the 1996 Pecos River kill (TPWD, unpublished data).

Threats to Other Aquatic Species

Although fish losses often get the most attention when these toxic events occur, entire aquatic ecosystems are threatened by golden alga. Prior to the 1985

TABLE 2. TPWD Management Survey Reports of Golden Alga Impacted Water Bodies.

Water Body	Fish Kills	Survey Dates	Species With Lowered Catch Rates, Poor Condition, or Both	Species Absent From Sampling Since (date)
Buffalo Springs	2003, 2006, 2008	2004-2005	Bluegill Channel catfish Largemouth bass White crappie	White bass Redbreast sunfish
Colorado City	2001-2002, 2004-2007	2002, 2004-2005	Largemouth bass	Bluegill (2002) Blue catfish (2000) Channel catfish (2000) Largemouth bass (2004) White bass (2000) White crappie (2000)
Diversion	2001-2002, 2004-2007	2004-2005	Bluegill Channel catfish Blue catfish Largemouth bass White bass White crappie	Striped bass (2001) Walleye (2001)
E.V. Spence	2001-2006, 2008	2007-2008	Bluegill Largemouth bass White bass White crappie	Blue catfish (2002) Channel catfish (2002) Striped bass (2002)
Granbury	2001, 2003-2007	2005-2006	Bluegill Largemouth bass Striped bass White crappie	
Kemp	2002-2003, 2005-2008	2004-2005	Bluegill Channel catfish Spotted bass Striped bass White bass White crappie	Blue catfish (2005)
Possum Kingdom	2001-2007	2006-2007	Blue catfish Largemouth bass Striped bass	Flathead catfish (2001) Smallmouth bass (2006) Spotted bass (2006)
Red Bluff	1985, 1988-2009, 2003, 2006-2007	2003-2005		No fish collected in gill net surveys (2003-2005)
Whitney	2001, 2003-2007	2007-2008	Blue catfish	Smallmouth bass (2005)

Note: TPWD, Texas Parks and Wildlife Department.

kill, the invasive Asiatic clam *Corbicula fluminea* was widespread in Pecos River waters with estimates of 100 Asiatic clams per square foot common in some areas (James and De La Cruz, 1989). A 1996 survey of the benthic macroinvertebrate community of the Pecos River below Red Bluff Reservoir reported a single Asiatic clam (Larson, 1996) and today this species persists but in low numbers (G. Larson, Texas Commission on Environmental Quality, personal communication). Of greater concern is the status of the Ouachita rock pocketbook *Arkansia wheeleri*, a mussel species that is federally listed as endangered and which occurs in Texas in tributaries of the Red River (Howells *et al.*, 1996).

Besides being considered a harmful algal bloom (HAB) species, *P. parvum* also has been characterized as an ecosystem disruptive algal bloom species (Sunda *et al.*, 2006) because of its ability to cause recurrent fish kills and affect various taxa and trophic levels within a water body. Other aquatic taxa that can be affected by golden alga toxins include crayfish, gilled amphibians, polychaetes, oligochaetes, gastropods, and some species of zooplankton (Sarig, 1971; James and De La Cruz, 1989; Igarashi *et al.*, 1999; Fistarol *et al.*, 2003) while aquatic insects appear unaffected (Shilo, 1972).

Effects on Public Freshwater Hatcheries

When golden alga first appeared in Texas waters, TPWD became concerned about this emerging threat to aquatic life. Matters worsened in 2001 when golden alga appeared in the source waters for two TPWD freshwater hatcheries, Possum Kingdom Reservoir near Graford and Diversion Lake near Electra,

that supply water to the Possum Kingdom and DSFH, respectively. The latter hatchery suffered unprecedented fish mortalities to *P. parvum* toxicity. Fish losses included over 14.2 million fry and fingerlings of striped bass and Palmetto bass (female striped bass × male white bass), 1,500 black basses, and thousands of channel catfish, rainbow trout *Oncorhynchus mykiss*, and koi *Cyprinus carpio*. Although *P. parvum* was identified as responsible for these fish kills at the hatchery, staff was not prepared to manage the problem. Besides the immediate loss of fingerlings for stocking public waters, the smallmouth bass and largemouth bass programs were set behind for years as the TPWD struggled to replace lost brood fish and equip staffs with the knowledge and tools required to manage golden alga to allow fish production.

To deal with the golden alga problem, the TPWD formed the hatchery *P. parvum* Task Force, composed of hatchery biologists, managers, and research personnel. Available information on golden alga and potential mitigation techniques were assembled and researched (Barkoh and Fries, 2005; Barkoh *et al.*, 2010). Through these efforts successful treatments were developed and implemented including the use of ammonium sulfate (Barkoh *et al.*, 2003, 2004; Smith, 2005a) and copper algaecides to kill cells (Dorzab and Barkoh, 2005a,b; Smith, 2005a), potassium permanganate to detoxify water (Dorzab and Barkoh, 2005a; Smith, 2005b), and ozone and UV sterilization to kill the alga or neutralize the toxin (Smith, 2005c). Efficacious treatments were formulated into management plans (Lyon *et al.*, 2005; Smith, 2005d) used by the affected hatcheries to control blooms and toxicity to successfully produce and rear fish.

Because the statewide distribution of golden alga was relatively unknown apart from locations where fish kills were confirmed, hatchery biologists were concerned that stocking activities could contribute to expanding the range of *P. parvum* in the state. Hatchery staff investigated and adopted efficacious methods to disinfect hauling units and minimize the incidental spread of golden alga (Barkoh *et al.*, 2010).

Effects on Local Economies

Estimated economic impacts of HAB in the U.S. exceed US$80 million annually (Jewett *et al.*, 2007; Hoagland *et al.*, 2008). This figure is generally attributed to coastal HAB species because the economic effects of inland HAB species are uncertain (Lopez *et al.*, 2008). Not only have golden alga kills had a significant effect on aquatic resources in Texas, but also considerable adverse economic effects have occurred in areas plagued by these toxic episodes. Oh and Ditton (2005, 2006) studied the economic impact of golden alga fish kills in terms of decreased water-based recreation and tourism on the three-county area surrounding Possum Kingdom Reservoir (Palo Pinto, Stephens, and Young Counties) after the 2001 and 2003 fish kills. Using tourism-derived data for 1986 to 2004, they estimated the economic losses associated with the 2001 and 2003 fish kills to be US$2.8 and US$1.1 million, respectively. These estimates do not include the value of the lost fish. Arguably, comparable economic losses have occurred in the surrounding communities of other affected water bodies and lead us to believe that the total effect of golden alga upon Texas resources and economies is far greater than it appears from the available data.

EFFORTS TO MANAGE THE GOLDEN ALGA PROBLEM

TPWD Strategies to Address the Problem

After the catastrophic fish kills at Possum Kingdom Reservoir in 2001, TPWD convened a workshop in August 2001 to discuss the golden alga problem. Representatives from local and state agencies, university researchers, elected officials, and community leaders attended and identified research needs and developed an action plan based on the available data. This information was incorporated in a report submitted to Texas legislators that outlined areas of potential research in early 2002 (http://www.tpwd.state.tx.us/landwater/water/environconcerns/hab/ga/legreport.phtml, *accessed* August 2009).

A webpage was created to provide information on blooms and toxic events in the state as well as to disseminate research findings (http://www.tpwd.state.tx.us/landwater/water/environconcerns/hab/ga/, *accessed* August 2009). The TPWD also established the Golden Alga Taskforce (GATF) whose mission is to protect aquatic ecosystems by understanding toxic golden alga blooms and to avoid, minimize, and mitigate their impacts through coordination and collaboration with stakeholders. The GATF organized a workshop held in October 2003 in Fort Worth, Texas that brought together resource managers, water authorities, interested stakeholders, and international and national experts to discuss the current knowledge of this toxic alga and actions needed to develop management options. The GATF used this information to establish research objectives and subsequently coordinated a variety of research projects including potential bloom and toxicity control techniques, bloom dynamics, toxin characterization, monitoring of

affected water bodies, and economic impacts caused by these harmful blooms. From 2003 to 2007, the TPWD expended more than US$2 million of both legislatively appropriated and U.S. Fish and Wildlife Service State Wildlife Grant funds to support internal and external research efforts aimed at understanding and ultimately controlling toxic golden alga blooms. Results from some of these studies appear in this issue (e.g., Barkoh *et al.*, 2010; Brooks *et al.*, 2010; Grover *et al.*, 2010; Hagström *et al.*, 2010; Kurten *et al.*, 2010; Lutz-Carrillo *et al.*, 2010; Roelke *et al.*, 2010; and Schug *et al.*, 2010) and otherwise may be found at http://www.tpwd.state.tx.us/landwater/water/environconcerns/hab/ga/research/research.phtml, *accessed* August 2009.

After six years of intensive research, a second international workshop was held in 2009 that brought together international, national, and statewide researchers and public stakeholders to share research findings, discuss ways to bridge the gap between research and management of the alga in reservoirs, and streamline efforts to better understand and hopefully manage this problematic species. The GATF plans to build on the accomplishments discussed at the workshop by guiding and funding relevant collaborative research aimed at developing management options for large water bodies.

Dissemination of information on golden alga to educate the public has been another important goal of the GATF. The GATF has prepared a fact sheet on golden alga for distribution to the public and "Guidelines for Golden Alga *P. parvum* Management Options for Ponds and Small Reservoirs (Public Waters) in Texas" (Sager *et al.*, 2007) for distribution statewide and also available online http://www.tpwd.state.tx.us/publications/pwdpubs/media/pwd_rp_t3200_1404.pdf, *accessed* August 2009. In addition to print information, the GATF organized or coordinated public meetings in municipalities near affected waters where TPWD personnel have lead discussions to explain the status of golden alga and current efforts to mitigate the problem.

CURRENT STATUS AND FUTURE ISSUES

Spread of Golden Alga in Texas

Toxic golden alga blooms are a recurring problem in many Texas inland water bodies and may have the potential to spread to other systems, exacerbating an already troublesome problem. Preliminary results from a statewide survey of golden alga distribution indicated *P. parvum* is widespread throughout Texas (TPWD, unpublished data). Although there are no known golden alga related kills in Texas coastal waters, a fish kill was reported in the Gulf Coast Water Authority Canal system near Texas City on Galveston Bay (less than 10 miles from the coast) in March 2006. Worldwide, golden alga is usually reported from brackish and estuarine waters, thus it is plausible that Texas coastal waters are vulnerable. In February 2009, golden alga was identified near Kingsville, Texas as responsible for a fish kill in a private stock pond less than 10 miles from the coast (Baffin Bay). TPWD biologists have been monitoring coastal bays for the presence of *P. parvum* and associated water quality parameters since 2007 in an effort to document conditions prior to any possible golden alga problems that may occur in the future.

Need for Cost-Effective, Viable Treatment Options

Although there are successful treatments for controlling golden alga in hatchery ponds and small water bodies, no effective management tool has been identified for larger water bodies. However, toxic blooms are seldom homogeneous throughout an entire reservoir and refuge areas could be created using treatments that are efficacious and cost-effective for small impoundments. Currently, copper-containing algaecides (e.g., Cutrine®-Plus, Applied Biochemists, Germantown, Wisconsin and copper sulfate) are the only EPA-approved treatments that are known to be effective against golden alga (Rodgers, 2010). In laboratory trials, products containing or producing hydrogen peroxide (e.g., GreenClean Pro, BioSafe Systems, Glastonbury, Connecticut and PAK 27, Solvay Chemicals, Houston, Texas) have been demonstrated to be effective against *P. parvum* cells and toxins (G. Southard, TPWD, unpublished data). The GATF plans to conduct demonstration studies aimed at minimizing bloom formation and to evaluate the effects of created refugia on recreational fisheries in select reservoirs in the near future.

SUMMARY

Since first documented in Texas waters in 1985, *P. parvum* remains a serious threat to Texas aquatic resources. Golden alga continues to bloom annually and cause fish mortality in private stock ponds, public water bodies, and two state freshwater hatcheries. These blooms affect fisheries, imperiled fish species, other aquatic life, recreation, and local economies. There has been progress in understanding the alga

and factors related to bloom formation and toxicity. Techniques have been developed to manage and mitigate its effects in hatchery ponds and smaller water bodies. However, there are still no cost-effective approaches to prevent these massive kills from occurring in large reservoirs popular for recreational use. The TPWD has provided leadership, coordination, and communication in dealing with the golden alga issue, not only in Texas but also at national and international levels.

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